Theoretical Skills

International Physics Olympiad · Lecture + Problems

Introductory study guide · 14 original problems

> Organized around the IPhO syllabus. This guide is not exhaustive; consult the syllabus study PDF and the IPhO syllabus webpage for the full scope and qualifications.

2.1 General

Model the system before calculating: identify scales, make justified approximations and exploit symmetry. Check dimensions and limiting cases.

2.2 Mechanics

Kinematics relates displacement, velocity and acceleration. Apply force and torque balance, Newton's laws, conservation of energy and momentum, and angular momentum. For gravitation use the potential $-GM/r$ and orbital energy; fluids require pressure, buoyancy, continuity and Bernoulli's equation. Surface tension adds an energy cost for area.

2.3 Electromagnetic fields

Use superposition, Coulomb's law and Gauss's law for symmetric charge distributions. Magnetic forces act perpendicular to velocity; induction follows Faraday's and Lenz's laws. In circuits, Kirchhoff's rules and Ohm's law determine currents; capacitor and inductor energies are $CV^2/2$ and $LI^2/2$. For sinusoidal AC, combine complex impedances.

2.4 Oscillations and waves

For a mass and spring, $\omega = \sqrt{(k/m)}$; near a stable equilibrium small oscillations are often harmonic. Waves satisfy $v = f\lambda$. Interference depends on path difference, diffraction limits resolution, and refraction obeys $n_1 \sin\theta_1 = n_2 \sin\theta_2$. Thin lenses obey $1/f = 1/u + 1/v$ with consistent signs.

2.5 Relativity

In inertial frames the spacetime interval is invariant. The Lorentz factor is $\gamma = 1/\sqrt{(1-v^2/c^2)}$; a moving clock's elapsed time is dilated by γ. Total energy and momentum satisfy $E^2 = p^2c^2 + m^2c^4$.

2.6 Quantum physics

Photons carry $E = hf$ and $p = h/\lambda$. Atomic transitions emit or absorb energy equal to the level difference. The uncertainty relation $\Delta x \Delta p \geq \hbar/2$ limits simultaneous localization and momentum

precision. Nuclear decay follows N = N₀e^(−λt), with half-life ln 2/λ.

2.7 Thermodynamics and statistical physics

The ideal gas law is pV = nRT. With work done by the gas W, the first law is ΔU = Q − W. An ideal reversible heat engine between temperatures T_h and T□ has efficiency 1 − T□/T_h. Radiation from a black body has power σAT^4; phase changes involve latent heat.

A physicist’s toolkit: why symmetry and limits matter

Imagine balancing two identical pulls. If nothing distinguishes left from right, the answer cannot choose one side. Symmetry can make components vanish before any algebra. A model is a simplified story, not reality itself: state what is neglected, then test the story in an extreme case. For example, if friction tends to zero, a result should approach the frictionless answer.

Mechanics: the accounting of motion

A force is not needed to keep an object moving; it changes momentum. Imagine keeping two separate books: momentum changes only by external impulse, while energy may move between kinetic, potential and thermal forms. Momentum can be conserved in a collision even when kinetic energy is not. For rotation, replace mass by moment of inertia and speed by angular speed; torque changes angular momentum.

Fields and circuits: local pushes, global bookkeeping

An electric field says what force a small positive test charge would feel at a location. Gauss’s law counts flux through a closed surface; it finds the field easily only when symmetry also tells us its direction and magnitude. A circuit is not a reservoir that uses up charge: charge flows around a loop, while energy is transferred to components. A capacitor stores energy in its electric field, and an inductor resists changes in current.

Waves: adding disturbances, not intensities

A wave transports a disturbance. When two coherent waves meet, add their displacements first: equal in-phase displacements reinforce, opposite ones cancel. Intensity depends on the square of the resulting amplitude. A standing wave is built from oppositely travelling waves; its nodes do not mean that each individual travelling wave has stopped.

Relativity: compare events before comparing clocks

An event has a time and a place. A clock measures proper time between events on its own worldline, but different observers disagree about simultaneity. Time dilation applies to a moving clock between two ticks, with both ticks at the same place in that clock’s frame. The invariant interval lets observers agree on the relationship between events even when their coordinates

differ.

Quantum and thermal physics: probabilities and energy flow

Quantum theory predicts probabilities for outcomes, not merely a tiny classical orbit that we have failed to see. The energy of a photon depends on frequency, not on intensity; increasing intensity at fixed frequency increases photon flux. Temperature is not heat: temperature describes thermal state, whereas heat is energy transferred because of a temperature difference. The first law is energy bookkeeping; the second law limits which transformations are possible.

Problems

Try each problem before opening the worked solutions. Exercises are original MAIS PAC teaching material, not official IPhO problems.

1. A body starts from rest and accelerates uniformly at 2 m/s² for 3 s. Find its speed and displacement.

2. A 2 kg mass moves at 3 m/s. Find its kinetic energy.

3. A 12 V ideal battery is connected across a 4 Ω resistor. Find current and power.

4. A wave travels at 12 m/s with frequency 3 Hz. Find its wavelength.

5. A thin converging lens has f = 10 cm and an object at u = 30 cm. Find the image distance.

6. A clock moves at 0.6c relative to a lab. How much lab time passes during 4 s of proper time?

7. A radioactive sample has a half-life of 2 hours. What fraction remains after 6 hours?

8. An ideal heat engine operates between 600 K and 300 K. What is its maximum efficiency?

9. Challenge 1 — A small block is released from rest at height h on a frictionless fixed track. At the bottom it collides and sticks to a stationary block of equal mass on a horizontal

frictionless surface. Find their common speed and the fraction of the original mechanical energy converted to internal energy in the collision.

10. Challenge 2 — A satellite is in a circular orbit of radius r around a spherical body of mass M. A brief tangential impulse reduces its speed to a fraction α of the original circular speed, where $0 < \alpha < 1$. Find the semimajor axis a of the new orbit and identify whether the impulse point is periapsis or apoapsis.

11. Challenge 3 — Two identical capacitors of capacitance C are isolated. One initially has voltage V; the other is uncharged. They are connected in parallel with matching polarities by idealized conducting wires. Find the final voltage and the energy dissipated during redistribution.

12. Challenge 4 — A source emits light of vacuum wavelength λ. In a double-slit experiment with slit separation d and screen distance $L \gg d$, determine the distance between neighbouring bright fringes near the centre. State the approximation.

13. Challenge 5 — A particle at rest of rest mass M decays into two photons travelling in opposite directions. Find each photon’s energy. What changes if the photon energies are unequal in the laboratory?

14. Challenge 6 — One mole of a monatomic ideal gas expands quasistatically and isothermally from V to 2V at temperature T. Find the work done by the gas and heat absorbed. Then compare its entropy change with that of a reservoir at the same T.